

Clinical and biochemical characterization of women with polycystic ovary syndrome in North Rhine-Westphalia.

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Polycystic ovary syndrome (PCOS), defined as the combination of oligoanovulation and hyperandrogenism, affects more than 5% of women of reproductive age. Insulin resistance and hyperinsulinemia appear to play an important role in its pathogenesis. Here, we will present a characterization of a PCOS cohort from North Rhine-Westphalia in Germany. Clinical features, family history as well as endocrine and metabolic parameters were prospectively recorded from 200 successive patients. All patients were evaluated for insulin resistance and beta-cell-function by oral glucose tolerance test. Patient data were compared with those of 98 age-matched control women. PCOS patients showed significantly higher BMI, body fat mass and androgen levels as well as impaired glucose and insulin metabolism. A positive family history of PCOS and diabetes was more frequent in PCOS patients. Insulin resistance (71%) was the most common metabolic abnormality in PCOS patients followed by obesity (52%) and dyslipidemia (46.3%), with an incidence of 31.5% for the metabolic syndrome. C-reactive protein and other cardiovascular risk factors were frequently elevated even in young PCOS patients. While the clinical characteristics and endocrine parameters of this German PCOS cohort were heterogeneous, they were comparable to those from other Caucasian populations.